

Exercise 1

TUMOR DETECTION AND
CLASSIFICATION USING SVM

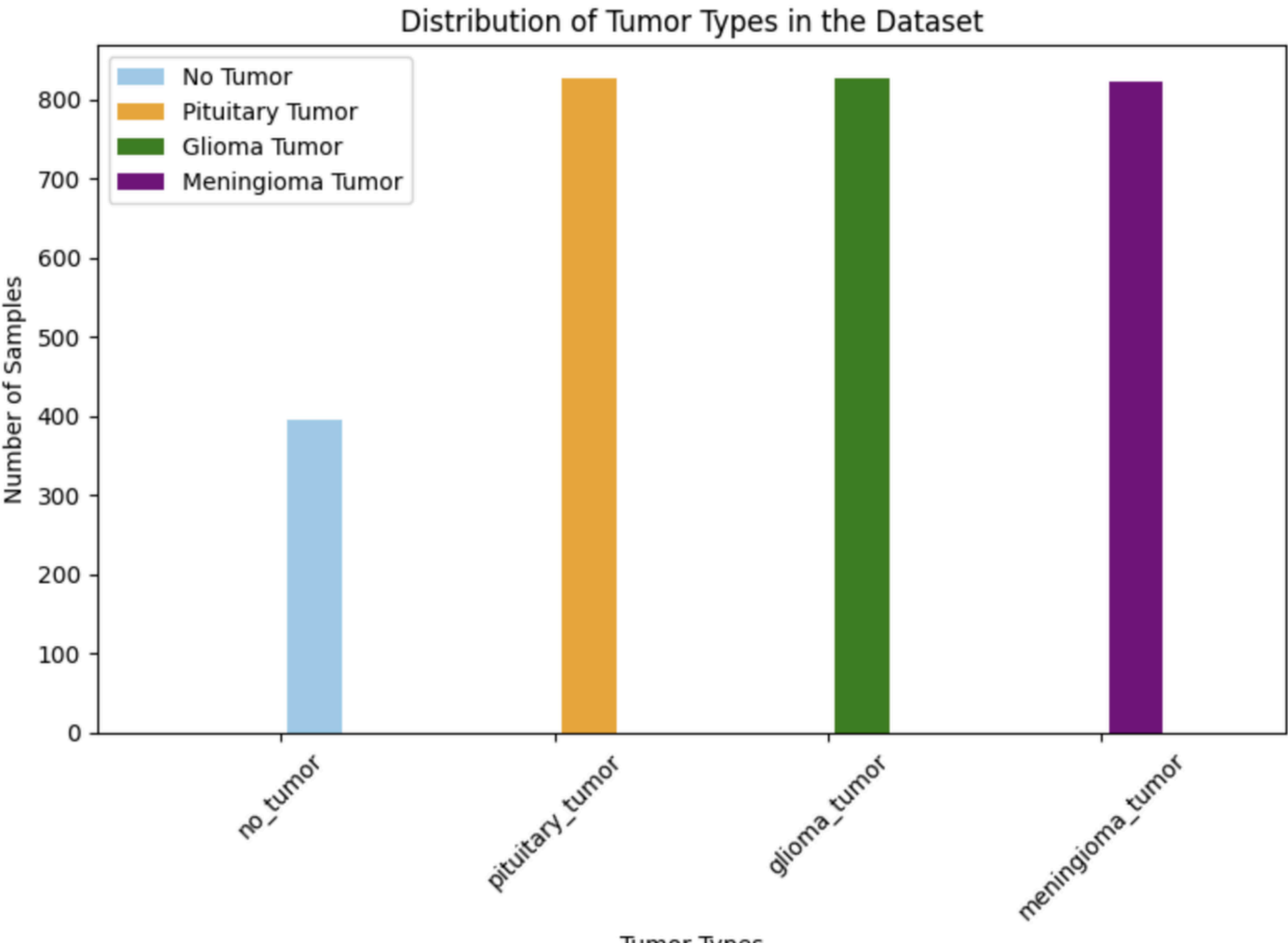
RESULTS (SVM-RBF KERNEL):

Training Score: 0.9651567944250871

Testing Score: 0.8310104529616724

Tumor Type Counts:

	Index	Tumor Type	Count
0	1	No Tumor	395
1	2	Pituitary Tumor	827
2	3	Glioma Tumor	826
3	4	Meningioma Tumor	822



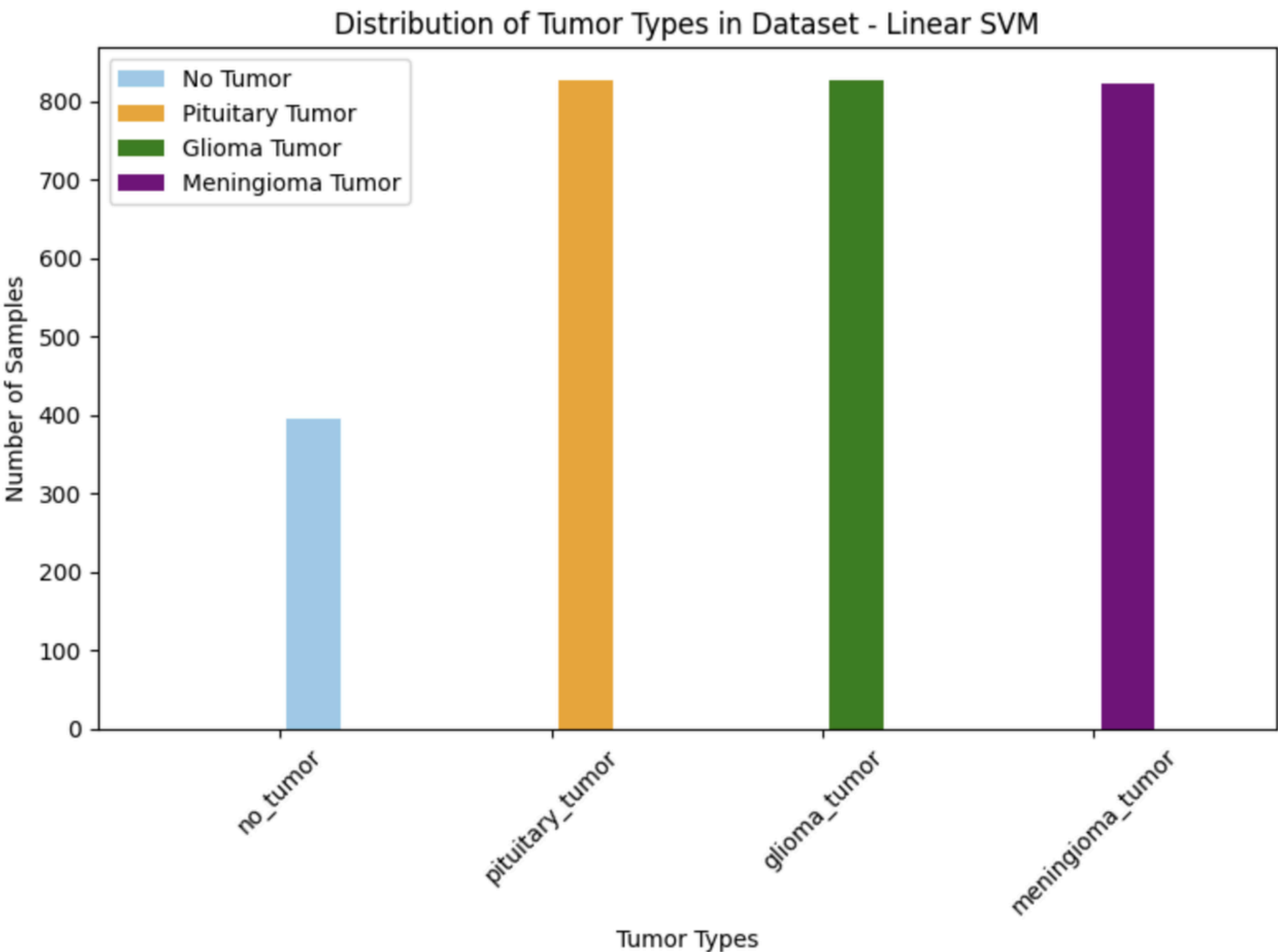
RESULTS (SVM-LINEAR):

Training Score: 1.0

Testing Score: 0.8013937282229965

Tumor Type Counts (Linear SVM Dataset)

	Index	Tumor Type	Count
0	1	No Tumor	395
1	2	Pituitary Tumor	827
2	3	Glioma Tumor	826
3	4	Meningioma Tumor	822



RESULTS: SVM(RBF)

(DIFFERENT TRAIN- TEST DATA SPLITTING)

20-80 Split

Training Score: 0.9529616724738676

Testing Score: 0.7321428571428571

40-60 Split

Training Score: 0.9581881533101045

Testing Score: 0.7874564459930313

60-40 Split

Training Score: 0.9576074332171893

Testing Score: 0.8440766550522648

80-20 Split

Training Score: 0.9668989547038328

Testing Score: 0.8432055749128919

RBF Kernel - 20% Training Data
Training Score: 0.9547038327526133
Testing Score: 0.7138501742160279

	Index	Tumor Type	Count
0	1	No Tumor	79
1	2	Pituitary Tumor	166
2	3	Glioma Tumor	165
3	4	Meningioma Tumor	164

RBF Kernel - 40% Training Data
Training Score: 0.9494773519163763
Testing Score: 0.778164924506388

	Index	Tumor Type	Count
0	1	No Tumor	158
1	2	Pituitary Tumor	331
2	3	Glioma Tumor	330
3	4	Meningioma Tumor	329

RBF Kernel - 60% Training Data
Training Score: 0.9610917537746806
Testing Score: 0.8179442508710801

	Index	Tumor Type	Count
0	1	No Tumor	237
1	2	Pituitary Tumor	496
2	3	Glioma Tumor	496
3	4	Meningioma Tumor	493

RBF Kernel - 80% Training Data
Training Score: 0.9625435540069687
Testing Score: 0.8327526132404182

	Index	Tumor Type	Count
0	1	No Tumor	316
1	2	Pituitary Tumor	662
2	3	Glioma Tumor	661
3	4	Meningioma Tumor	657

PERFORMANCE ANALYSIS

RBF Kernel - 20% Training Data

Accuracy: 0.7138501742160279

F1-Score: 0.6951669974380358

Sensitivity: 0.6893215094355657

Specificity: 0.901571122557361

RBF Kernel - 40% Training Data

Accuracy: 0.778164924506388

F1-Score: 0.7716576036199883

Sensitivity: 0.763325331523043

Specificity: 0.9243616994195059

RBF Kernel - 60% Training Data

Accuracy: 0.8179442508710801

F1-Score: 0.8148838627684649

Sensitivity: 0.8101113320858716

Specificity: 0.9383622454684961

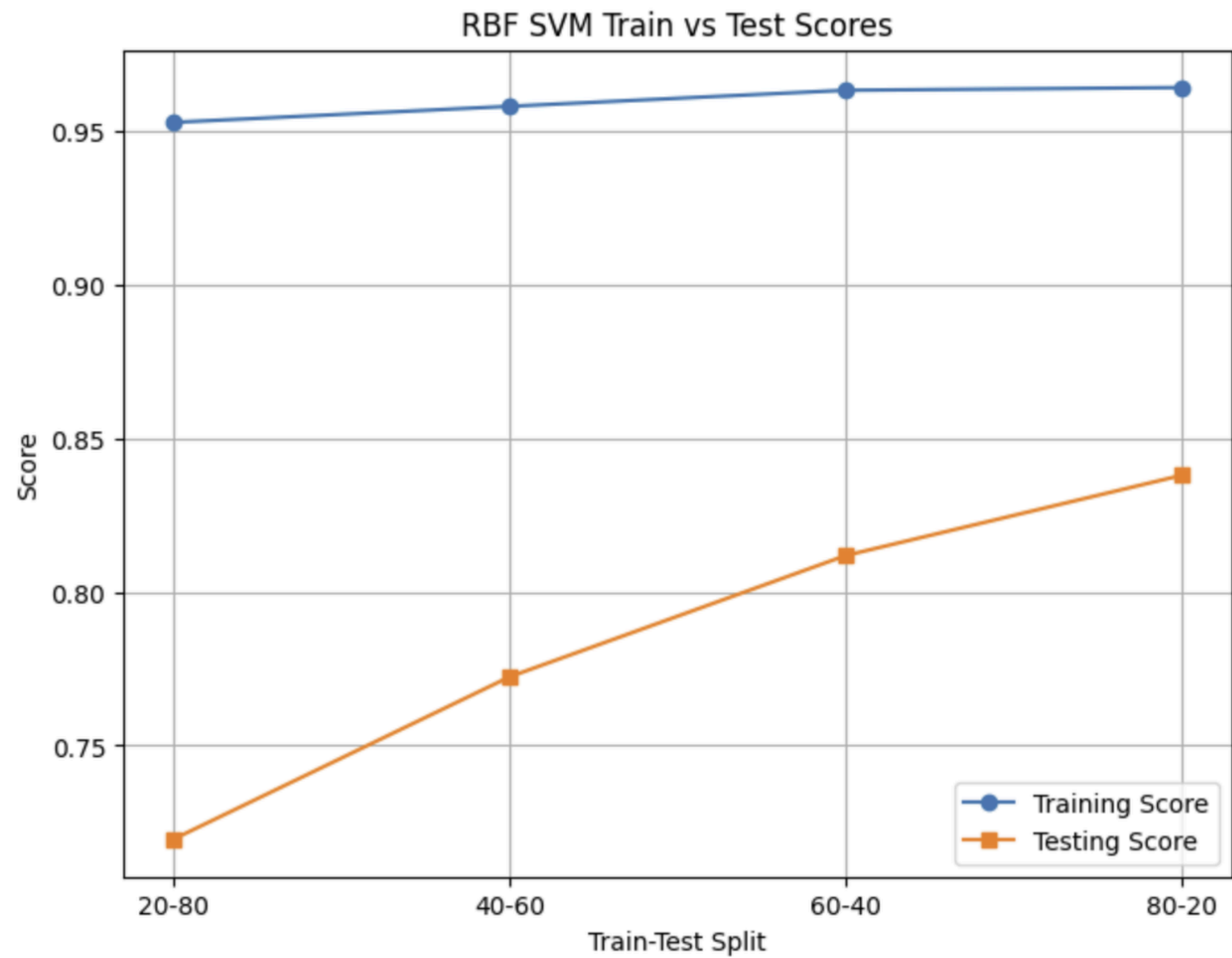
RBF Kernel - 80% Training Data

Accuracy: 0.8327526132404182

F1-Score: 0.8288886796761112

Sensitivity: 0.8215573456079786

Specificity: 0.9435504186115432



RESULTS: (SVM LINEAR)

(DIFFERENT TRAIN- TEST DATA SPLITTING)

20-80 Split

Training Score: 1.0

Testing Score: 0.7221254355400697

40-60 Split

Training Score: 1.0

Testing Score: 0.7619047619047619

60-40 Split

Training Score: 1.0

Testing Score: 0.7700348432055749

80-20 Split

Training Score: 1.0

Testing Score: 0.8013937282229965

Tumor Type Counts - 20-80 Split

	Index	Tumor Type	Count
0	1	No Tumor	78
1	2	Pituitary Tumor	148
2	3	Glioma Tumor	183
3	4	Meningioma Tumor	165

Tumor Type Counts - 40-60 Split

	Index	Tumor Type	Count
0	1	No Tumor	144
1	2	Pituitary Tumor	324
2	3	Glioma Tumor	352
3	4	Meningioma Tumor	328

Tumor Type Counts - 60-40 Split

	Index	Tumor Type	Count
0	1	No Tumor	221
1	2	Pituitary Tumor	489
2	3	Glioma Tumor	514
3	4	Meningioma Tumor	498

Tumor Type Counts - 80-20 Split

	Index	Tumor Type	Count
0	1	No Tumor	304
1	2	Pituitary Tumor	665
2	3	Glioma Tumor	681
3	4	Meningioma Tumor	646

PERFORMANCE ANALYSIS

Linear Kernel - 20% Training Data

Accuracy: 0.7290940766550522

F1-Score: 0.7243899682330085

Sensitivity: 0.7151243723025037

Specificity: 0.9077475658897677

Linear Kernel - 40% Training Data

Accuracy: 0.772938443670151

F1-Score: 0.7687522412741843

Sensitivity: 0.7626907886399797

Specificity: 0.9226095390939819

Linear Kernel - 60% Training Data

Accuracy: 0.7952961672473867

F1-Score: 0.7935016343665066

Sensitivity: 0.7846666891253031

Specificity: 0.9303730124792632

Linear Kernel - 80% Training Data

Accuracy: 0.8275261324041812

F1-Score: 0.8256782197054756

Sensitivity: 0.8120636747219026

Specificity: 0.9407609098318144

